

Metric Spaces

General Topology

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Definition

The Cartesian product $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$ of n copies of $\mathbb{R}$ is known as the **n -dimensional Euclidean space**. An element of $\mathbb{R}^n$ is denoted by $(x_1, \dots, x_n)$ or simply x . The numbers x_i are called its **components** or **coordinates**. Zero-dimensional Euclidean space $\mathbb{R}^0$ is, by convention, the singleton $\{0\}$.

Remark

It is a well-known fact from linear algebra that $\mathbb{R}^n$ is an n -dimensional real vector space with the usual operations of scalar multiplication and vector addition (defined componentwise), and, in fact, it is the only one (in the sense that any other n -dimensional linear space is isomorphic to $\mathbb{R}^n$).

Definition

We refer to an element of $\mathbb{R}^n$ either as a **point** or as a **vector**, depending on whether we wish to emphasise its location or its direction and magnitude. The geometric properties of $\mathbb{R}^n$ are derived from the **Euclidean inner product**, defined by

$$\langle x|y \rangle \equiv x \cdot y = x_i y_i, \quad i = 1, 2, \dots, n.$$

In particular, the **norm** or **length** of a vector $x \in \mathbb{R}^n$ is given by

$$||x|| \equiv |x| = \sqrt{\langle x|x \rangle}.$$

Definition

If x and y are nonzero vectors in $\mathbb{R}^n$, the **angle between x and y** is defined via

$$\cos \theta \doteq \frac{\langle x|y \rangle}{|x||y|}.$$

Further, given two points $x, y \in \mathbb{R}^n$, the **line segment from x to y** is the set

$$\{x + t(y - x) : 0 \leq t \leq 1\},$$

and the **distance between x and y** is $|x - y|$. A **closed ray in $\mathbb{R}^n$** is any set of the form $\{x + t(y - x) : t \geq 0\}$ for two distinct points $x, y \in \mathbb{R}^n$, and the corresponding **open ray** is $\{x + t(y - x) : t > 0\}$.

Definition

A map $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ between subsets of Euclidean spaces is said to be **continuous at $x \in U$** if for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $y \in U$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$. Such a map is said to be **continuous** if it is continuous at each point of its domain U .

Definition

A **sequence** in $\mathbb{R}^n$ is any map $X : \mathbb{N} \rightarrow \mathbb{R}^n$. The value of this map at some given n is written as $X(n) \equiv x_n$, and we will frequently use the abbreviation $X \equiv (x_n) \equiv (x_i)$ to denote the sequence itself. A sequence (x_i) of points in $\mathbb{R}^n$ is said to **converge** to $x \in \mathbb{R}^n$ if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|x_i - x| < \varepsilon \quad \text{for } i \geq N.$$

A sequence is **divergent** if it does not converge to any point, and **bounded** if there is some $R \in \mathbb{R}$ such that $|x_i| \leq R$ for all i .

Definition

Suppose M is any set. A **metric on M** is a function $d : M \times M \rightarrow \mathbb{R}$ satisfying the following properties for all $x, y, z \in M$:

- i) **(Symmetry)** $d(x, y) = d(y, x)$.
- ii) **(Positivity)** $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
- iii) **(Triangle inequality)** $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (M, d) is called a **metric space**. When the metric is understood from the context, one often just says “ M is a metric space”.

Example 1.1

Prove that $d : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $d(x, y) \doteq |x - y|$ is a metric on any subset U of $\mathbb{R}^n$.

Definition

For any $x \in M$ and $r > 0$, the **(open) ball of radius r around x** is the set

$$B_r(x) = \{y \in M : d(y, x) < r\},$$

and the **closed ball of radius r around x** is

$$\overline{B}_r(x) = \{y \in M : d(y, x) \leq r\}.$$

Definition

A subset $A \subseteq M$ of a metric space M is said to be an **open subset of M** if it contains an open ball around each of its points, and it is said to be a **closed subset of M** if $M \setminus A$ is open.

Theorem 1.2 (Properties of Open Subsets of a Metric Space)

Let M be a metric space. Then

- i) both M and $\emptyset$ are open subsets of M ;*
- ii) the intersection of finitely many open subsets of M is an open subset of M .*
- iii) the union of arbitrarily many open subsets of M is an open subset of M .*

Definition

Suppose M is a metric space and let A be a subset thereof. We say that A is **bounded** if there exists a positive number R such that $d(x, y) \leq R$ for all $x, y \in A$. If A is nonempty and bounded, we define the **diameter of A** to be the number $\text{diam } A = \sup_{x, y \in A} \{d(x, y)\}$.

Definition

If (M_1, d_1) and (M_2, d_2) are metric spaces and x is a point in M_1 , a map $f : M_1 \rightarrow M_2$ is said to be **continuous at x** if for any $\varepsilon > 0$ there exists $\delta > 0$ such that $d_1(x, y) < \delta$ implies $d_2(f(x), f(y)) < \varepsilon$ for all $y \in M_1$; and f is said to be **continuous** if it is continuous at each point in M_1 .

Definition

A **sequence** in M is any map $X : \mathbb{N} \rightarrow M$. A sequence (x_i) of points in a metric space M is said to **converge** to $x \in M$ if for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$d(x_i, x) < \varepsilon \quad \text{for } i \geq N.$$

In this case, we call x the **limit of the sequence** and write

$$x_n \rightarrow x \quad \text{or} \quad \lim(x_n) = x.$$

A sequence is **divergent** if it does not converge to any point. A sequence (x_n) in a metric space is said to be **bounded** if its image $\{x_n\}_{n \in \mathbb{N}}$ is a bounded subset of M .

Definition

A sequence (x_n) in a metric space is said to be **Cauchy** (or a **Cauchy sequence**) if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \quad \text{for } n, m \geq N.$$

A metric space in which every Cauchy sequence converges to a value admitted in the space is said to be **complete**.

Theorem 1.3 (Open Subset Criterion for Continuity)

A map $f : M_1 \rightarrow M_2$ between metric spaces is continuous if and only if the preimage of every open subset is open.